

Circuits (Capacitance and Resistance)

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1 Introduction

Basic Text List:

- **Generator:** Motors, anything that can create a potential Differences
- **Resistors:** Convert electrical energy to heat
- **Capacitors:** Store charge

2 Important Notice about Batteries

- Batteries do NOT supply charges (electrons come from capacitors / wires).
- Battery = *charge pump* = source of energy

3 Capacitors (plate capacitors)

Capacitors have the ability to store charge over a potential difference. Capacitance is calculated as follows:

$$C = \frac{q}{V}$$

Where q is the charge that the conductor has and the V is the potential difference across the capacitor. Capacitance also comes with a new unit, the **Farad**:

$$\frac{[C]}{[V]} = \frac{[J]}{[C]} = [F]$$

3.1 Calculating For Capacitance

Capacitance depends on the following:

- Geometry of the plates
- Medium between plates (usually we're going to use air, though, we can also use other mediums)

To solve, follow the following steps

1. Find the electric field between plates
2. Find the ΔV from the field
3. Apply that to the capacitance equation $C = \frac{q}{V}$

3.2 Calculating the Capacitance between Parallel Plates

3.2.1 Finding the Electrical Field between two plates

With a area of A and a distance between of d , with the medium between them air (shown by the ϵ_0)

$$\begin{aligned}\Phi &= EA = \frac{q_{\text{enc}}}{\epsilon_0} = \frac{q}{A\epsilon_0} \\ &= \frac{\sigma}{\epsilon_0}\end{aligned}$$

3.2.2 Finding voltage from field

$$\Delta V = - \int \vec{E} \delta \vec{s} = \int_{-\text{plate}}^{+\text{plate}} \vec{E} \delta \vec{s} = E \int \delta s = Ed$$

We removed the negative sign because while the field is going from the positive plate to the negative plate, the direction vector is doing the opposite. We also double the field because of the two plates.

3.2.3 Solving for capacitance

$$C = \frac{q}{V} = \frac{q}{\frac{q}{A\epsilon_0} \times d} = \frac{A\epsilon_0}{d}$$

3.3 Calculating for the Capacitance between Cylindrical Capacitors

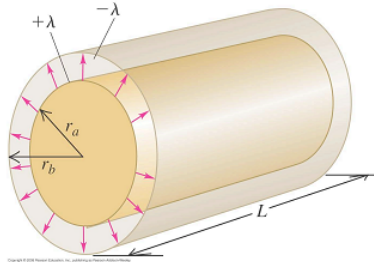


Figure 1: Examples of a Cylindrical Capacitor

3.3.1 Find Electrical Field between Two of the Cylinders

$$\begin{aligned}\Phi &= EA = \frac{q_{\text{enc}}}{\epsilon_0} \\ A &= 2\pi rL \\ E &= \frac{q}{2\pi\epsilon_0 rL}\end{aligned}$$

3.3.2 Find Voltage from Field

$$\Delta v = - \int \vec{E} \delta \vec{s} = - \int_b^a \frac{q}{2\pi r L \epsilon_0} \delta r = \int_a^b \frac{q}{2\pi r L \epsilon_0} \delta r = \frac{q}{2\pi L \epsilon_0} \ln \frac{b}{a}$$

3.3.3 Solving for the Equation

$$C = \frac{q}{v} = \boxed{\frac{2\pi L \epsilon_0}{\ln \frac{b}{a}}}$$

4 Capacitors in Parallel vs Circuits in Series

Always remember the formula $C = \frac{Q}{V}$

4.1 Capacitors in Parallel

If a capacitor is in parallel, you can think it as adding up metal plates. Voltages are the same, and charge is proportionate to the capacitance at the capacitor. For an example 3 capacitors in series, the,

$$\begin{aligned} q_T &= V_b \Sigma C \\ q_T &= q_1 + q_2 + q_3 \\ V_1 &= V_2 = V_3 = V_B \\ C_T = C_{eq} &= \frac{q_T}{V_T} = \frac{q_1}{V_1} + \frac{q_2}{V_2} + \frac{q_3}{V_3} \\ C_{eq} &= \Sigma C \end{aligned}$$

Note: This is opposite than resistors.

4.2 Capacitors in Series

If a capacitor is in series, you can think about it as adding up the air gaps. In the end, it's one big capacitor. Thus charge is equal.

$$\begin{aligned} q_T &= q_1 = q_2 = q_3 \\ V_T &= V_1 + V_2 + V_3 \\ \frac{1}{C_{eq}} &= \frac{V_T}{q_T} = \frac{V_1 + V_2 + V_3}{q} = \frac{V_1}{q_1} + \frac{V_2}{q_2} + \frac{V_3}{q_3} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \end{aligned}$$

5 Capacitance and Energy

One would think that since $q = CV_0$, and since $U = qV$, capacitance is CV^2 . However, this is not true, since the voltage of a graph changes over time.

$$\begin{aligned} \Delta U &= \int dq \delta U = \int V_{delta} q = \int \frac{q}{c} \delta q \\ &= \frac{q^2}{2c} \end{aligned}$$

6 Energy Density

Energy density is literally just the energy divided by the volume.

$$u = \frac{U}{\text{vol}} = \frac{\epsilon_0 V^2}{2d^2} = \frac{1}{2} \epsilon E$$